

Data-driven design of Nb-W refractory alloys using Transformer-based stress-strain modeling

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ABSTRACT

Generative AI provides a powerful route for predicting material behavior in data-scarce regimes, an especially pressing challenge for refractory alloys where extreme processing environments and high melting points hinder large-scale data collection. Here, we introduce a Transformer-based generative framework that models the entire deformation trajectory rather than focusing solely on isolated scalar properties (such as YS, UTS), by synthesizing full stress-strain curves of Nb-W-based refractory alloys. Applied to Nb521-derived systems, the model captures the sequential evolution of tensile deformation and accurately predicts elastic-plastic responses across a wide compositional and processing space. The framework demonstrates utility in three downstream tasks: (i) inverse design of composition and processing, yielding >35% strength improvement without loss of ductility; (ii) interpretable regime analysis via a Mixture-of-Experts (MoE) mechanism, highlighting the beneficial roles of Hf and Ta in ductility, consistent with experimental and DFT validation; and (iii) seamless integration of generated curves into finite element simulations for structural performance assessment. Together, these results establish a coherent generative-analytic-predictive workflow for mechanism-aware, data-efficient design of advanced refractory alloys.

1. Introduction

Refractory alloys, particularly Nb- and W-based systems, are critical structural materials for next-generation aerospace, nuclear, and hypersonic applications due to their exceptional high-temperature strength and thermal stability [1–6]. Despite decades of development, achieving a favorable balance between strength and ductility remains a major challenge [7–9]. These multicomponent solid solutions, often further strengthened by interstitial dopants such as C, N, and B, present a vast design space where alloying, processing, and microstructural evolution are strongly interdependent [10–13]. Their mechanical responses are inherently path-dependent, governed by evolving internal states such as dislocation accumulation, phase transformation, and damage localization [14–17]. Consequently, accurate performance prediction requires modeling the entire deformation trajectory rather than focusing solely on isolated scalar properties.

Among the various descriptors of mechanical performance, the

tensile stress-strain curve is particularly valuable because it captures not only scalar quantities such as yield strength, ultimate tensile strength, and ductility, but also the evolution of work hardening and deformation history. In Nb-W-based refractory alloys, these responses arise from a combination of intrinsic mechanisms (e.g., solid-solution strengthening, phase stability) and extrinsic microstructural features (e.g., grain size, precipitate morphology) that depend strongly on thermomechanical history [18–20]. These mechanisms are highly coupled and path-dependent, and conventional scalar-property regression cannot capture their full complexity, making Nb-W alloys an ideal system to demonstrate the advantages of a generative stress-strain modeling framework. To establish a practical benchmark for model development, we focus on the widely used Nb521 alloy (Nb-5W-2Mo-1Zr-0.07C, wt %). In the fully recrystallized condition, Nb521 shows a tensile strength of ~500 MPa at room temperature and >300 MPa at 1200 °C [21,22]. While these values demonstrate good thermal stability, the alloy's limited strength-ductility balance highlights the need for systematic

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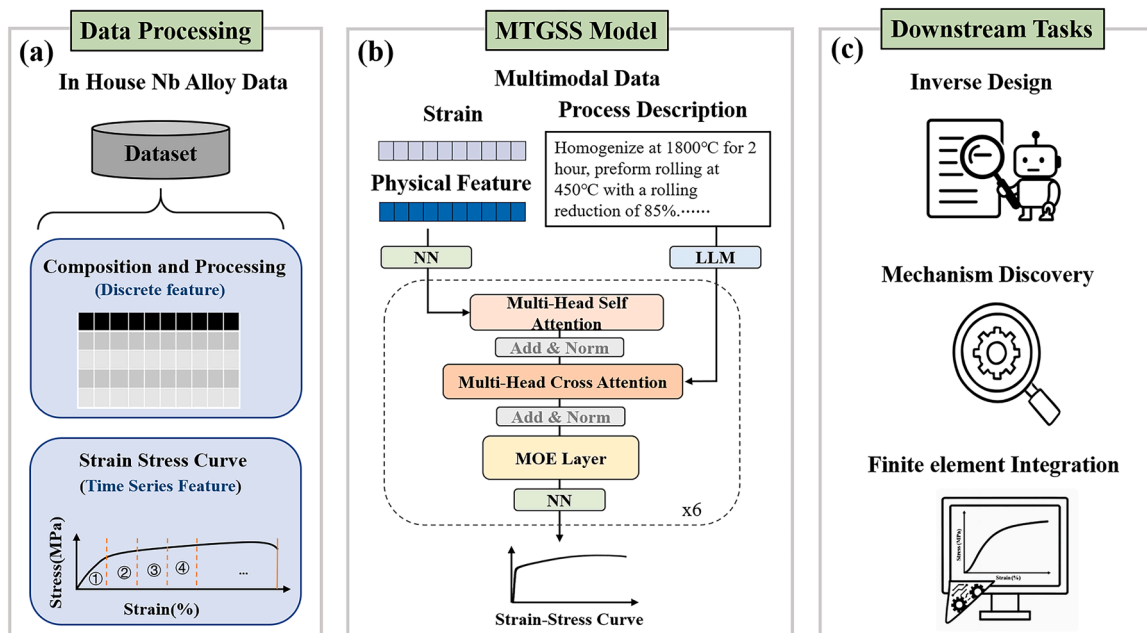


Fig. 1. Schematic overview of the Transformer-enabled generative modeling framework for constitutive behavior prediction of refractory alloys. The workflow consists of three main modules: (a) data collection, including literature-sourced RHEA data and in-house Nb-based alloy datasets represented as discrete stress-strain curves; (b) a multimodal Transformer architecture that takes strain fragments and physics-informed descriptors as forward numerical inputs, and alloy/process textual prompts encoded by SteelBERT as side inputs, with feature fusion via multi-head cross-attention and regime-specific processing via a Mixture of Experts (MoE) layer; and (c) downstream tasks including inverse design, mechanism discovery, and finite element integration.

optimization through both compositional tuning and process control.

Despite decades of progress in constitutive modeling, capturing the full mechanical response of materials across diverse chemistries and processing regimes remains challenging. Traditional physically based approaches—including empirical flow rules [23–26], crystal plasticity theory [27–30], and multiscale simulations such as molecular dynamics [31–33]—provide mechanistic insight but often require extensive parameter calibration, rely on idealized assumptions (e.g., microstructural homogeneity or simplified loading paths), and lack transferability across systems. In response, data-driven surrogate models have been proposed to accelerate property prediction by learning from existing datasets [34–39]. However, most are formulated as supervised regression tasks targeting a limited set of scalar outputs (e.g., yield strength, Young’s modulus), and thus cannot capture the sequential, history-sensitive nature of full-field deformation responses. Moreover, these models often generalize poorly to unseen composition-process regimes. Several recent efforts have applied sequence models (such as RNN, LSTM) to the full stress-strain curve prediction. Yet many rely on synthetic datasets generated from simplified constitutive models (e.g., Ramberg-Osgood, Johnson-Cook), which do not reflect the nonlinear, path-dependent complexities observed in experimental deformation behavior [40–43]. As a result, such models often miss critical features—such as yield point phenomena, transient hardening, or post-necking softening—and are limited in their utility for performance prediction or alloy optimization.

To overcome the limitations of traditional constitutive models and largely supervised-based ML approaches, we develop a generative modeling framework based on the Transformer architecture for producing full-field stress-strain curves of Nb-W-based refractory alloys. Trained on experimentally measured data, the model learns to generate continuous, path-dependent deformation trajectories conditioned on alloy composition and processing parameters. Within this alloy family, we demonstrate three representative applications: (i) inverse design of Nb-W alloys with enhanced strength–ductility balance, (ii) interpretable identification of elemental effects (Ta, Hf, W) on strain hardening and ductility, and (iii) finite element integration of generated constitutive

curves for structural performance assessment. Together, these results establish a mechanism-aware and data-efficient framework tailored to Nb-W alloys. By leveraging full stress–strain trajectory generation, this approach captures path-dependent deformation features, advancing the methodological frontier of machine learning in constitutive modeling while providing practical guidance for the design of refractory alloys with optimized strength–ductility synergy.

2. Methodology

2.1. Multimodal-transformer-enabled constitutive model

As shown in Fig. 1(a), our multimodal-Transformer-enabled constitutive modeling framework integrates three core components: (i) data preprocessing and representation, (ii) a multimodal Transformer-based generative model for strain-stress curve (hereafter referred to as MTGSS), and (iii) downstream applications for alloy design and analysis.

2.1.1. Data collection and representation

The training dataset consists of 92 tensile stress-strain curves collected from two sources: 8 curves were obtained from published literature on refractory high-entropy alloys (RHEAs), and 84 were generated from our group’s experimental studies on Nb-based refractory alloys. The data encompass a range of compositions involving the elements Nb, W, Mo, Zr, Hf, Ti, Ta, Si, V, Sc, Ru, B, Y, C, and N. These curves span a wide range of compositions and thermomechanical processing conditions, and each is stored as a discrete array of paired strain-stress values, typically up to 2 % strain, with 512 uniformly sampled points.

To complement the mechanical response data, we compute a set of physically meaningful alloy descriptors for each composition, including valence electron concentration (VEC), mixing enthalpy, mixing entropy, and atomic size mismatch. These features help embed domain knowledge into the learning process and improve the model’s generalizability.

2.1.2. Multimodal transformer architecture

The core of our framework is a multimodal Transformer-based

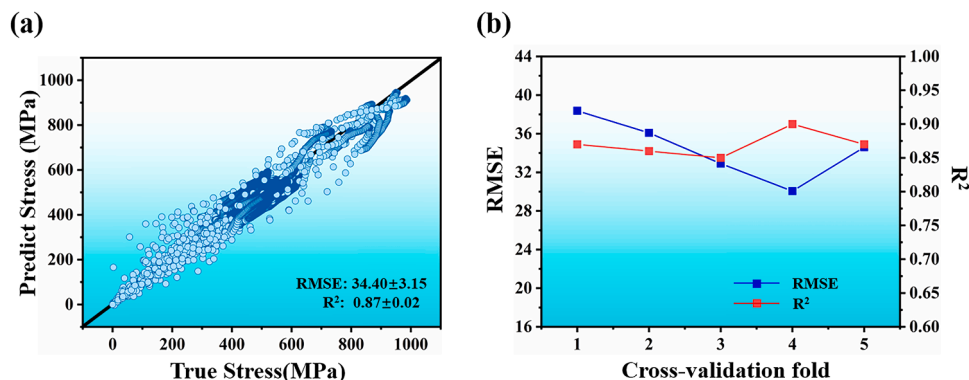


Fig. 2. Extrapolative performance of MTGSS in pointwise stress prediction across unseen alloy compositions, evaluated via Nb-W-stratified five-fold cross-validation. (a) Aggregated correlation between predicted and ground-truth stress values across all five held-out folds, where each point denotes a stress value at a discretized strain level; the diagonal line ($y = x$) indicates perfect agreement. (b) Fold-wise performance summary (RMSE and R^2) for the five composition strata defined by Nb and W contents, demonstrating stable extrapolative accuracy across compositionally separated folds.

generative model that predicts stress-strain sequences in an autoregressive fashion. The MTGSS model takes two streams of forward numerical input:

- (1) the strain fragment, representing the incremental strain history up to each prediction step (512 points up to 2 % strain), and
- (2) the physics-informed alloy descriptors, such as VEC, enthalpy, entropy, and atomic radius mismatch. These two input types are separately encoded using fully connected neural networks and concatenated to form the input embedding. The resulting representation is processed through a multi-head self-attention mechanism, enabling the model to learn temporal and physical correlations within the data.

In parallel, the MTGSS model receives a side input in the form of a natural language description of the alloy's composition and processing route. We design a standardized prompt template to unify descriptions across samples (e.g., “Homogenize at 1800 °C for 2 hours, perform rolling at 450 °C with a rolling reduction of 85%, ...”). These textual prompts are encoded using SteelBERT [44], a pre-trained domain-specific language model tailored for metallurgical language, producing semantic embeddings that are injected into the Transformer via a multi-head cross-attention mechanism. This enables the model to align numerical features with contextual knowledge, such as thermomechanical processing effects or alloying strategies.

To account for the distinct mechanical regimes present in stress-strain curves—namely elastic deformation, yielding, and plastic flow—we incorporate a Mixture of Experts (MoE) layer into the Transformer architecture. In the present work, 6 experts of sub-networks are adopted to capture the shape of the strain-stress response. Each expert is trained to specialize in a particular deformation regime, and the model dynamically selects the most 2 appropriate experts for each segment of the curve. This mechanism enhances both predictive accuracy and physical interpretability by allocating model capacity to regime-specific response features.

As illustrated in Fig. 1(b), the core model architecture consists of a stack of six Transformer blocks, each comprising a series of key sub-modules. The block begins with a multi-head self-attention mechanism, which captures both temporal dependencies within the strain sequence and correlations among the physics-informed descriptors. This is followed by a multi-head cross-attention module that integrates contextual information from the textual embeddings, allowing the model to align semantic knowledge from alloy composition and processing with the numerical features. To ensure stable optimization and effective gradient flow, each attention layer is equipped with residual connections followed by layer normalization (Add & Norm). Finally, a MoE layer is

incorporated at the end of each block to dynamically allocate computational resources to specialized expert subnetworks, each of which focuses on modeling a specific mechanical regime—such as elastic, yielding, or plastic deformation. This modular design enhances both prediction accuracy and interpretability by enabling the model to adaptively specialize across different stages of the stress-strain response.

The MTGSS model was trained to autoregressively generate the stress value at each strain step by minimizing a composite loss function that combines mean squared error, Kullback-Leibler divergence, and a curvature penalty, thereby balancing numerical accuracy and physical smoothness. Unlike traditional regression models that only predict scalar properties, our framework outputs full stress-strain trajectories, enabling rich, path-aware modeling of mechanical behavior under tension. Model architecture, training parameters, and the loss function are detailed in Supplementary Note 1 (Supporting Information).

2.1.3. Downstream tasks

In downstream applications, this generative framework supports three representative tasks: (1) Inverse design, where the model is used to search for alloy/process combinations that achieve target mechanical properties; (2) Mechanism discovery, by analyzing the learned attention patterns and expert activations to identify which elements (e.g., Hf, Ta) contribute to enhanced room-temperature ductility, and (3) Finite element integration, where the generated stress-strain curves are directly used in structural simulations for performance evaluation.

Together, this multimodal Transformer framework enables a unified, data-efficient, and mechanism-aware approach to modeling and designing next-generation refractory alloys. The complete details of the dataset, the multimodal-transformer-enabled constitutive model, as well as the code for the implementation, have been uploaded to <https://github.com/weidaokuo/MTGSS/tree/master>.

2.2. Experimental methodology

Nb-W-based refractory alloys were synthesized from high-purity elemental granules ($\geq 99.95\%$) via vacuum arc melting in a water-cooled copper crucible under argon atmosphere. Each ingot was remelted at least six times to ensure compositional homogeneity. The as-cast alloys were then homogenized at 1800 °C for 2 h, followed by furnace cooling, and subjected to rolling at 450 °C with an 85 % thickness reduction. Post-deformation annealing was performed at 1450 °C for 1 hour for composition optimization, while selected alternative annealing temperatures and durations were used in process optimization studies. All stress-strain data were extracted from samples tested after these thermal treatments.

Tensile tests were conducted using an Instron 5969 testing machine,

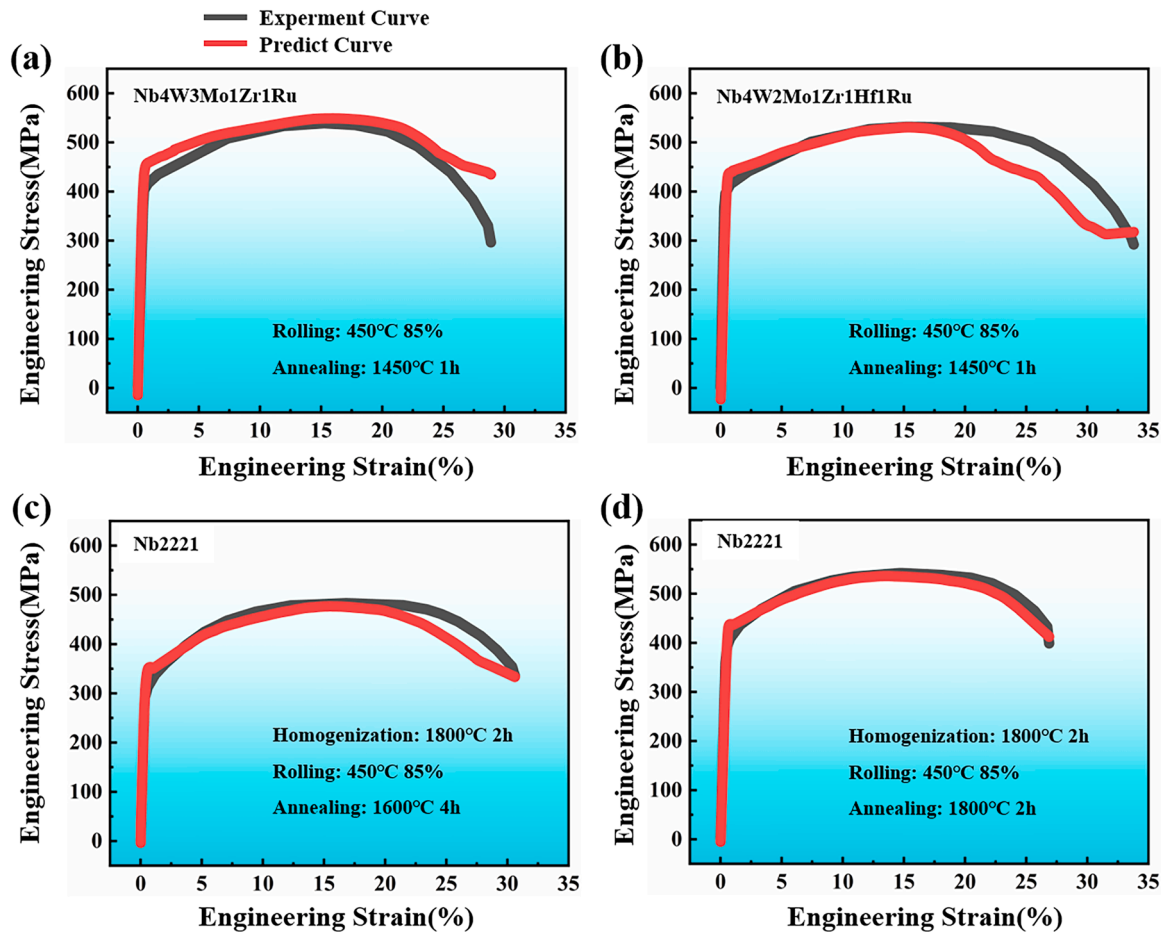


Fig. 3. Full-curve validation of the MTGSS model for representative Nb-W-based alloys. (a) and (b) Predicted and experimentally measured stress-strain curves for two different alloy compositions processed under identical conditions (85 % warm rolling, 1450 °C annealing). (c) and (d) Model prediction for the Nb2221 (Nb2W2Mo2Ta 1Hf0.5Zr0.05Co0.06Y0.04Si0.05B) alloy under two distinct heat treatments. The predicted curves accurately reflect process-induced changes in yield strength, hardening behavior, and softening at large strain, confirming the model's sensitivity to thermal history and its microstructural effects.

at a constant strain rate of $1 \times 10^{-3} \text{ s}^{-1}$. Specimens were machined using electrical discharge machining (EDM), with gauge sections of $8 \times 2 \times 1.5 \text{ mm}^3$, and polished to 2000 grit. An extensometer was used for accurate strain measurement. Microstructural characterization was conducted using a Thermo Scientific Talos F200X operated at 200 kV, employing transmission electron microscopy (TEM), dark-field (DF) imaging, and energy-dispersive X-ray spectroscopy (EDS). Thin TEM foils were first mechanically thinned to a thickness of $\sim 50 \text{ }\mu\text{m}$ and subsequently subjected to twin-jet electropolishing using a Struers Tenupol-5. Electropolishing was carried out at an applied voltage of $\sim 20 \text{ V}$ and a temperature of $-20 \text{ }^\circ\text{C}$, with an electrolyte consisting of 80 vol % methanol, 14 vol % sulfuric acid, and 6 vol % hydrofluoric acid.

3. Results and discussion

3.1. Validation of predictive performance

To systematically assess the predictive performance and extrapolative generalization of the proposed MTGSS model, we conducted a stringent pointwise evaluation of predicted versus reference stress values using a composition-stratified five-fold cross-validation protocol on the Nb-W-based refractory alloy dataset. Each stress-strain curve was discretized into stress values at uniformly sampled strain levels, enabling direct pointwise comparisons at identical strain steps. To enforce a genuine extrapolation setting, the dataset was partitioned into five composition strata according to Nb and W contents (19, 19, 18, 18, and 18 samples), and five-fold cross-validation was performed by

holding out one stratum for testing while training on the remaining four. This design prevents any test-fold compositions (and their close Nb-W neighborhoods under the stratification) from appearing in the corresponding training folds, thereby providing a faithful assessment of out-of-distribution extrapolation to unseen alloy chemistries.

Fig. 2(a) shows the aggregated correlation between predicted and ground-truth stresses over all five held-out folds, where each point corresponds to a single stress value at a given discretized strain level under a specific composition and processing condition; the diagonal ($y = x$) indicates perfect agreement. The data cluster tightly around the diagonal, evidencing strong quantitative consistency under composition-disjoint testing. Averaged across folds, MTGSS attains an RMSE of $34.40 \pm 3.15 \text{ MPa}$ and an R^2 of 0.87 ± 0.02 , demonstrating robust predictive fidelity in an extrapolative regime. Fig. 2(b) further reports fold-wise RMSE and R^2 , revealing consistently competitive performance across all strata and indicating that the model's accuracy is not confined to a particular region of the Nb-W compositional space. Overall, these results suggest that MTGSS effectively encodes composition-dependent effects together with processing information embedded in textual descriptors, enabling reliable prediction of mechanical responses under out-of-distribution conditions.

To further verify the model's ability to capture the full deformation trajectory, Fig. 3 presents representative comparisons between the predicted and experimentally measured stress-strain curves under controlled test scenarios. The evaluation focuses on how well the model reproduces both the overall curve shape and key deformation features under varying composition or processing conditions. In Fig. 3 (upper

Table 1

Process parameters of Nb-W alloys used in the composition optimization stage.

Composition (wt. %)	Homogenization parameters		Rolling parameters		Annealing parameters	
	T(°C)	Time (h)	T(°C)	Deformation (%)	T(°C)	Time (h)
Nb4W3Mo1Zr1Ru	1800	2	450	85	1450	1
Nb4W2Mo1Zr1Hf1Ru	1800	2	450	85	1450	1

panel), we examine two different Nb-W-based alloys with different compositions (listed in Table 1), all subjected to the same thermo-mechanical process: 85 % hot rolling, followed by annealing at 1450 °C for 1 hour. Despite the differences in the composition, the model accurately reconstructs the entire stress-strain trajectories, including the linear elastic regime, yield onset, and post-yield strain hardening. The predicted curves closely follow the experimental references, with minimal deviation in both slope and peak stress, demonstrating the model's ability to differentiate and learn compositional effects under fixed processing.

In Fig. 3 (lower panel), two different annealing conditions are applied to the Nb2221 alloys after homogenization and warm rolling: (i) annealing at 1600 °C for 4 hour, (ii) annealing at 1800 °C for 2 hour. The predicted stress-strain curves correctly reflect the process-induced changes in mechanical response, including differences in yield strength and strain hardening rate. Notably, the model captures subtle shape variations such as softening behavior at high strain, which can arise from recovery or microstructural coarsening effects at higher annealing temperatures. Further validation of the model's generalization is provided in Supplementary Note 2.

These results demonstrate that our MTGSS model can accurately reproduce full-field stress-strain curves across diverse alloy compositions and thermomechanical processes. Compared with prior models that rely on synthetic data or struggle with noisy experimental inputs, our framework achieves superior accuracy by integrating rich composition descriptors, physics-informed features, and natural language prompts—all of which enhance the fidelity of the input representation. More importantly, the Transformer's autoregressive and segmental prediction structure, augmented with the MoE module [45–48], enables the model to dynamically capture deformation regime transitions such as yield onset, strain hardening, and post-necking softening. Each

predicted stress segment is not only physically consistent but also accurately reflects the history-dependent mechanical response at specific stress or strain states. This fine-grained modeling capability distinguishes our approach from traditional sequence models which often fail to resolve localized inflection points [49].

3.2. Inverse design of Nb-W alloys with enhanced mechanical properties

To explore the compositional and processing design space for Nb-W-based alloys, we integrate our MTGSS model with an active learning-driven inverse design loop. The objective is to identify alloy candidates that exceed the mechanical performance of currently known compositions, as quantified by their full stress-strain curves. As illustrated in Fig. 4, the design loop operates as follows: in each iteration, a batch of candidate compositions and processing routes is sampled from a predefined design space. For each candidate, the MTGSS model generates a full stress-strain trajectory. To evaluate the improvement of mechanical performance, we define an increment of the strength-plasticity product (ΔSP) metric A, calculated as the integral of the difference between the predicted stress-strain curve $\sigma_{pred}(\epsilon)$ and the best-so-far experimental curve $\sigma_{ref}(\epsilon)$ within the dataset:

$$A = \int_0^{\epsilon_{max}} [\sigma_{pred}(\epsilon) - \sigma_{ref}(\epsilon)] d\epsilon$$

Here, ϵ_{max} denotes the maximum evaluated strain (typically corresponding to fracture, and $\epsilon_{max} = 0.2$ in the present study). A positive value of A indicates that the generated curve lies above the benchmark, implying potential improvement in both strength and ductility, while a negative value suggests no improvement over the best-so-far. This metric is particularly sensitive to key features such as yield strength, strain hardening, and post-yield behavior captured by the stress-strain response. Candidates are ranked according to their A-values, and the top two are selected for experimental validation. Their results are then incorporated into the training set to refine the MTGSS model. This closed-loop active learning strategy accelerates discovery by guiding exploration toward high-potential regions of the design space, while continuously improving the model's predictive capability. It is noted that our active learning workflow based on the MTGSS model is capable of quantifying uncertainty (see Supplementary Note 3). However, given

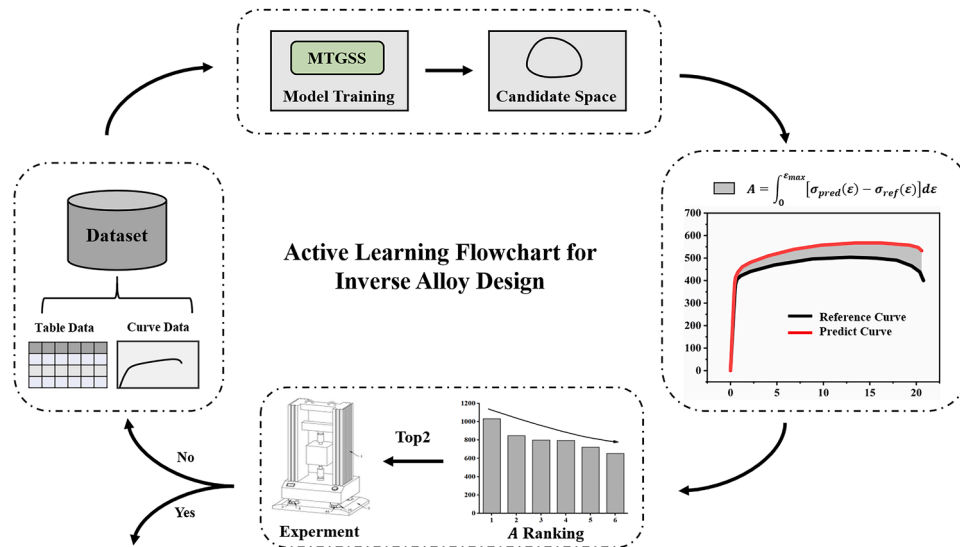


Fig. 4. Inverse design workflow integrating the MTGSS generative model with active learning. Each design iteration begins with candidate sampling from a predefined compositional and processing design space. For each candidate, the MTGSS model generates a full stress-strain trajectory, which is evaluated using an area-based performance metric A. Top-ranked candidates are selected for experimental validation, and the new data is incorporated into the training set to update the MTGSS model.

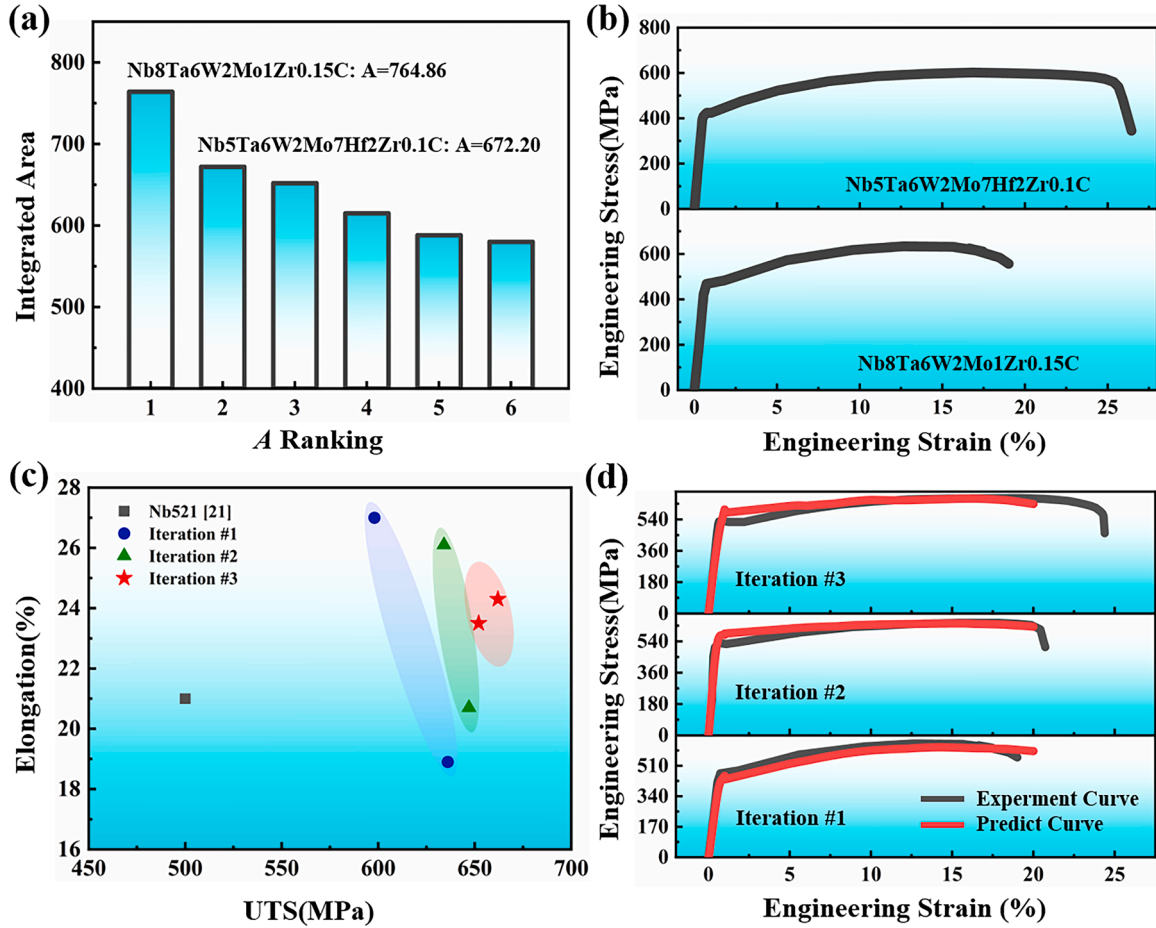


Fig. 5. Inverse design progression of Nb-W-based refractory alloys using the MTGSS-active learning framework. (a) Bar chart of predicted area-based performance metric A for the top six candidate alloys from the first design round. (b) Experimentally measured tensile stress-strain curves for the first-round candidates. (c) Strength-ductility map summarizing all validated alloys across three active learning iterations. The data points reveal a clear upward shift of the mechanical property frontier, indicating iterative improvement in both strength and ductility. (d) Evolution of validated stress-strain curves across successive design rounds.

the relatively low uncertainty exhibited by our model, this aspect is not considered further in the present study.

To demonstrate the application of the MTGSS-active learning framework for composition design, we conducted a case study targeting Nb-W-based alloys. The compositional space for the Nb-W alloys was

restricted to minor alloying variations around the commercial Nb521 alloy—a high-performance refractory alloy known for its excellent high-temperature strength and room-temperature workability [21,22]. As shown in Fig. 5(a), the first round of model-guided exploration produced a ranked bar chart of predicted A -values for the top six candidate alloys.

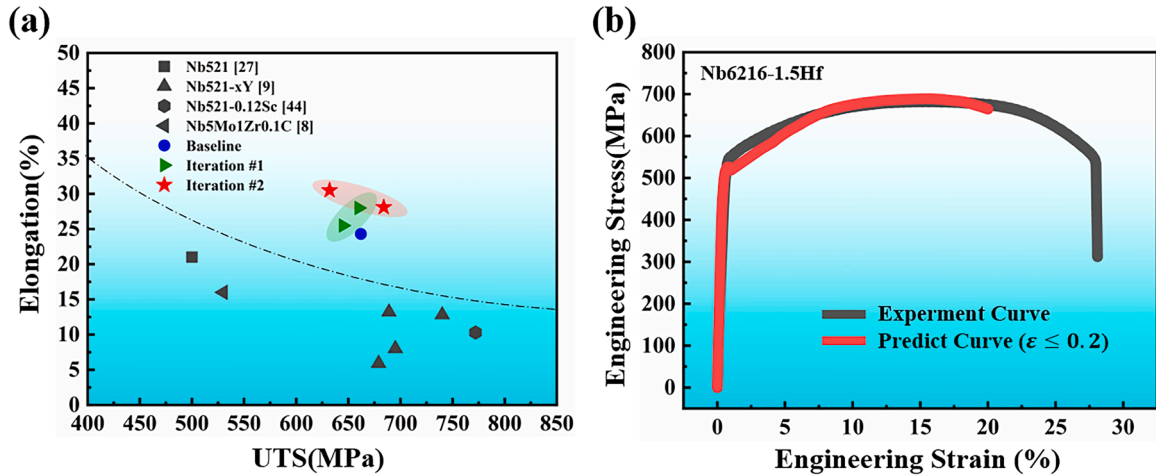


Fig. 6. Processing optimization of Nb6216-1.5Hf alloy using the MTGSS-active learning framework. (a) Strength-ductility map showing the evolution of tensile performance for Nb6216-1.5Hf under different thermomechanical treatment routes across two active learning iterations. (b) Stress-strain response of the optimized Nb6216-1.5Hf alloy, exhibiting a yield strength of 560 MPa, ultimate tensile strength of 684 MPa, and elongation to failure of 28 %.

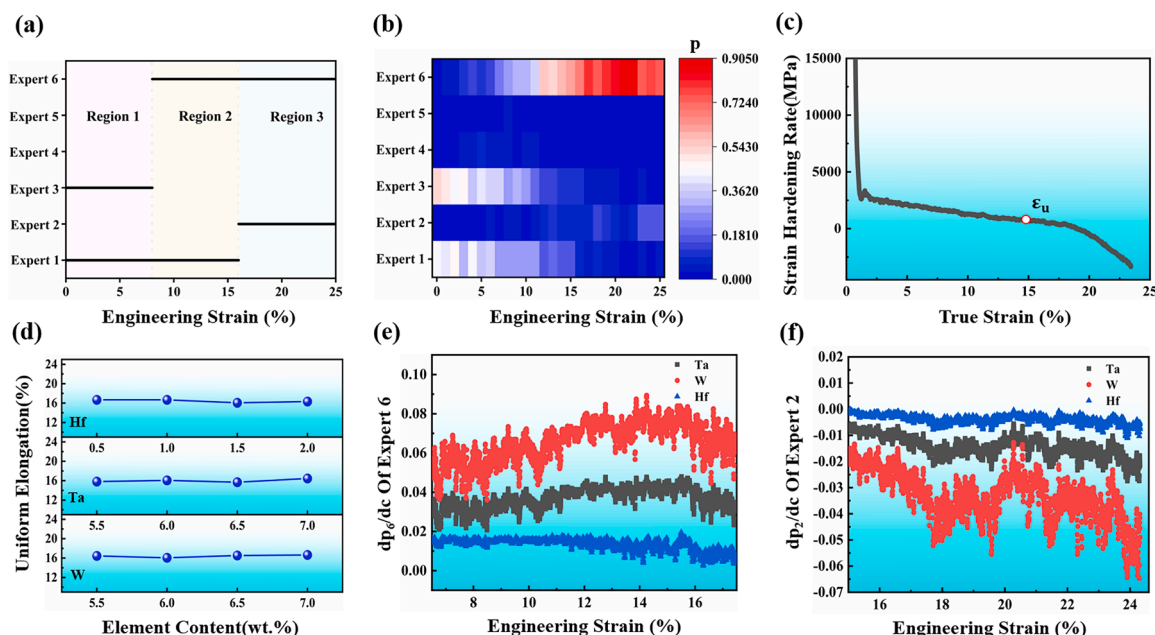


Fig. 7. Interpretable modeling of alloying effects via Mixture-of-Experts (MoE) activations in the MTGSS framework. (a) Strain-resolved expert activation patterns for the Nb6216-1.5Hf alloy. (b) Heatmap of expert activation probabilities across strain steps confirms the stage-specific behavior, with a sharp increase in activation probability P_2 for Expert 2 after $\epsilon_u = 16\%$. (c) Work hardening rate of the Nb6216-1.5Hf alloy. (d) Uniform elongation remains nearly constant ($\sim 16\%$) across variations in Ta, W, and Hf contents. (e) Strain-dependent sensitivity of Expert 6 activations, dP_6/dc , revealing its role of capturing work hardening. (f) Strain-dependent sensitivity of Expert 2 activations dP_2/dc reveals that W causes a rapid decline in activation—indicative of impaired ductility—while Ta slows this decline and Hf shows an intermediate effect, consistent with their respective roles in plasticity enhancement.

Based on this ranking, two high-ranking compositions were selected for experimental validation. The experimentally measured tensile stress-strain curves for these two candidates are displayed in Fig. 5(b). The alloys achieved a room-temperature ultimate tensile strength of 636 MPa and an elongation to failure of 18.9 %, validating the model's ability to generate compositionally optimized candidates with improved mechanical performance. To track the design progress over multiple rounds, Fig. 5(c) presents a strength-ductility map summarizing the mechanical performance of all candidate alloys validated during the active learning iterations. Each data point represents a performance landmark after one design loop. The property frontier initially rises sharply but shows progressively smaller gains with subsequent iterations.

After three iterative rounds of model-driven design and experimental validation, the optimized alloy composition Nb-6W-2Mo-1Zr-6Ta-1.5Hf-0.05C-0.2N (wt %)—was identified and designated as Nb6216-1.5Hf. This alloy exhibited a $\sim 32\%$ increase in tensile strength relative to the baseline Nb521 alloy, while maintaining comparable ductility. Fig. 5(d) illustrates the evolution of the validated stress-strain curves over successive design iterations, highlighting the progressive enhancement in mechanical performance. These findings demonstrate the efficacy of the MTGSS-enabled inverse design framework in finding Nb-W-based refractory alloys with superior mechanical properties, even under limited data constraints.

Building on the composition identified in the previous section—Nb6216-1.5Hf alloy—we further explored process optimization using the same MTGSS-active learning framework. The goal was to refine thermomechanical treatment routes to maximize mechanical performance at room temperature. As shown in Fig. 6(a), we constructed a strength-ductility map tracking the evolution of tensile properties over two active learning iterations. Each point represents a performance landmark corresponding to an experimentally validated processing condition. Similar to the composition design stage, the mechanical performance exhibited a consistent upward trend throughout the process optimization loop. After two iterative design cycles, the processing

route yielded a significantly enhanced tensile profile, as shown in Fig. 6 (b). The final alloy achieved a yield strength of 543 MPa, an ultimate tensile strength of 684 MPa, and an impressive elongation to failure of 28 %. Compared to the baseline performance of commercial Nb521 alloys, this represents an improvement of over 35 % in the UTS without sacrificing ductility, underscoring the effectiveness of our generative modeling framework in guiding multi-dimensional materials optimization.

3.3. Interpretable analysis of alloying effects

To gain insight into the influence of individual alloying elements on mechanical behavior, we leverage the Mixture-of-Experts (MoE) architecture embedded in the MTGSS model. In this setup, each expert sub-network is implicitly specialized to model a distinct deformation regime—such as elastic response, yield initiation, strain hardening, or plastic softening. During stress-strain curve generation, the model dynamically allocates varying attention weights to these experts depending on the strain state and input features. By tracking the expert activation patterns across different alloys and strain intervals, we obtain a fine-grained, stage-wise understanding of how different alloying elements affect mechanical response.

Fig. 7 illustrates the expert activation dynamics and their interpretability in relation to deformation stages and elemental contributions. As shown in Fig. 7(a), the MTGSS model consistently activates a subset of six expert sub-networks across the strain range. Experts 1, 3, and 6 dominate the early-stage deformation ($\epsilon < 16\%$), corresponding to elastic response and uniform plastic elongation, while Experts 2 and 6 become increasingly important beyond this regime. Expert 2 exhibits negligible activity prior to uniform elongation but rises sharply thereafter, capturing strain localization and post-necking softening. Expert 6, in contrast, maintains strong and persistent activation across both uniform and post-uniform regimes, reflecting its responsibility for strain hardening governed by dislocation interactions. This complementary behavior is further corroborated by the expert activation probability

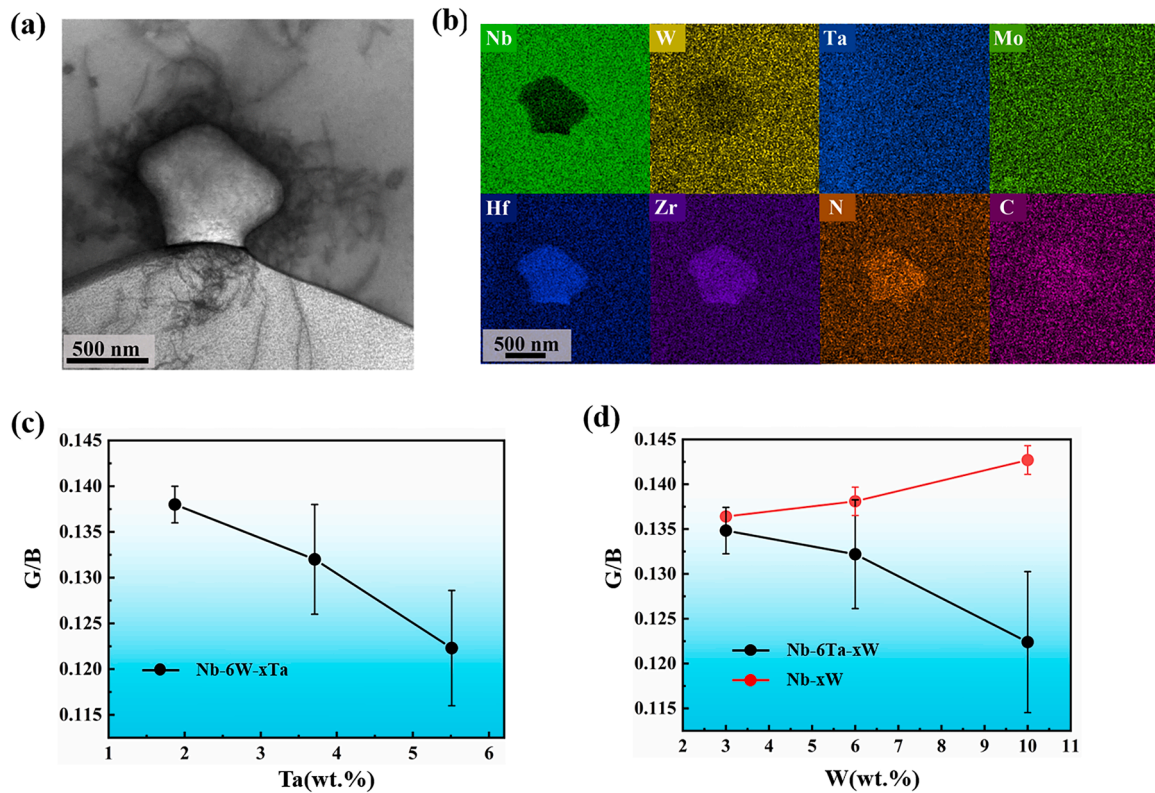


Fig. 8. Experimental and computational validation of MoE-based interpretability in Nb-W-based alloy design. (a) TEM micrograph of the Nb6216–1.5Hf alloy. (b) Corresponding EDS elemental maps reveal that Hf predominantly exists in the form of precipitates, whereas Ta and W are uniformly distributed as solid solution elements. (c) DFT-predicted Pugh ratios (or G/B ratios) for Nb-6W-xTa alloys. (d) The Pugh ratio trends as a function of W content for Nb-xW and Nb-6Ta-xW alloys.

heatmap in Fig. 7(b), which shows P_2 rising markedly around $\varepsilon \approx 16\%$ and dominating late-stage deformation, while P_6 persists as a secondary but significant contributor associated with sustained hardening. Together, this separation provides a physically meaningful decomposition of deformation behavior: Expert 6 encodes dislocation-interaction-driven hardening, whereas Expert 2 captures strain softening and failure processes.

This mechanistic assignment is further supported by the sensitivity analysis in Figs. 7(e) and 7(f). As shown in Fig. 7(e), dP_6/dc remains positive across the entire strain range for all three elemental variants, with a characteristic rise followed by a gradual decrease. This behavior confirms that Expert 6 consistently represents dislocation-interaction-driven hardening, which dominates during uniform plasticity and continues, albeit with decreasing intensity, into the post-necking regime. Building on this baseline role of Expert 6, Fig. 7(f) highlights how elemental variations primarily manifest through Expert 2. W induces the steepest decline in dP_2/dc with increasing strain, indicating that W suppresses post-necking plasticity and accelerates strain localization. In contrast, Ta shows the slowest decline, followed closely by Hf, implying that these elements enhance the ability to sustain large plastic strains beyond uniform elongation.

To validate the mechanistic insights derived from the MoE activations, we further performed microstructural characterization and first-principles calculations on the optimized Nb6216–1.5Hf alloy. Fig. 8(a) presents a representative TEM image of the alloy in its solution-treated and fractured state, revealing a refined, defect-rich microstructure. The corresponding EDS elemental maps in Fig. 8(b) show that while W and Ta are uniformly distributed within the matrix—indicating their primary role as solid solution strengtheners—Hf is predominantly enriched in discrete, nanoprecipitate-like regions. This observation confirms the dual strengthening behavior inferred from the MTGSS model: W and Ta contribute to the continuous deformation regime, while Hf contributes

to post-uniform ductility through second-phase interactions. The observed Hf-rich precipitates are consistent with prior reports on refractory alloy systems, where Hf has been shown to scavenge light-element impurities such as oxygen from grain boundaries [50,51]. These impurities, especially O, are known to promote intergranular embrittlement and severely limit ductility in Nb-based alloys. Hf reacts with such impurities to form dispersed HfC or HfO₂ particles, thereby purifying the grain boundaries and enhancing ductile fracture resistance [51]. The presence of such precipitates in Fig. 8 supports this known purification mechanism and reinforces the ductility-enhancing role of Hf suggested by the expert activations.

To further quantify the influence of solid solution elements on intrinsic plasticity, we conducted density functional theory (DFT) calculations of the Pugh ratio (i.e., G/B), a commonly used proxy for brittle-to-ductile transitions. As shown in Fig. 8(c), the Pugh ratio of Nb-6W-xTa alloys systematically decreases with increasing Ta content, confirming that Ta promotes ductility by enhancing lattice compliance. This trend is in excellent agreement with the MoE-based interpretation that Ta slows down the decline of Expert 2 activation and extends post-necking plastic flow. Interestingly, Fig. 8(d) reveals a more nuanced role for W. In binary Nb-xW alloys, the Pugh ratio increases with W content, indicating that excessive W addition tends to embrittle the alloy. However, in the ternary Nb-6Ta-xW system, the Pugh ratio decreases with W content, suggesting that the presence of Ta modulates W's influence and partially mitigates its embrittling tendency. Together, these experimental and computational findings corroborate the MoE-derived explanations of elemental roles in deformation, lending strong support to the reliability and interpretability of the generative model's internal representations.

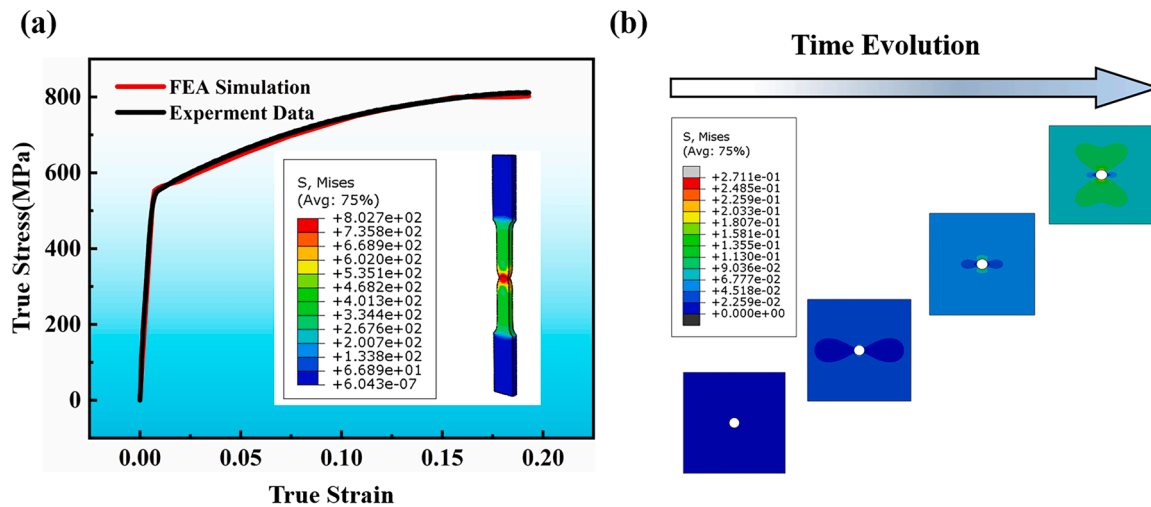


Fig. 9. Finite element validation of MTGSS-generated constitutive model using the Nb6216–1.5Hf alloy. (a) Comparison between the predicted and experimental true stress-strain curves under uniaxial tension. The inset displays the geometry and dimensions of the flat dog-bone tensile specimen used consistently in both simulation and physical testing. (b) Von Mises stress evolution in a Nb6216–1.5Hf thin sheet containing a central circular hole under increasing uniaxial strain, highlighting the progressive development of stress concentration and localization.

3.4. Structural performance evaluation via finite element integration

To assess the structural applicability of the constitutive model generated by our MTGSS framework, we incorporated the predicted stress-strain curves into a finite element (FEM) simulation pipeline. As shown in Fig. 9(a), the uniaxial tensile response of a flat dog-bone specimen was simulated using the predicted curve of the Nb6216–1.5Hf alloy. The FEM-predicted stress-strain response (red) exhibits excellent agreement with the experimental data (black), accurately capturing the elastic modulus, yield point, and strain hardening characteristics. The inset illustrates the geometry and dimensions of the specimen, ensuring consistency between computational inputs and physical testing configurations.

Fig. 9(b) further demonstrates the reliability of the MTGSS-derived constitutive model by simulating the deformation behavior of a thin Nb6216–1.5Hf sheet with a central circular hole under uniaxial tension. The evolution of von Mises stress contours at various strain stages reveals the progressive development of stress localization. Initially, stress is uniformly distributed, but with increasing strain, it begins to concentrate symmetrically along the transverse axis, forming a butterfly-shaped stress field centered at the hole. This localization intensifies during the plastic regime, eventually extending along both transverse and longitudinal directions.

Such simulations highlight the model's ability to not only predict global mechanical performance but also capture intricate local stress redistribution in defect-bearing geometries. This capability is especially crucial for the structural integrity assessment of components operating in extreme environments, where pre-existing micro-defects and geometric discontinuities are common. Together, Figs. 9(a) and 9(b) validate that our generative framework serves as a bridge between materials-level design and component-level structural analysis.

4. Conclusions

In summary, we developed a multimodal generative modeling framework based on the Transformer architecture to predict full-field tensile stress-strain responses of Nb-W-based refractory alloys. By integrating structured numerical descriptors, unstructured compositional and processing text prompts, and autoregressive sequence learning, the model effectively captures the nonlinear, path-dependent mechanical behavior of complex alloys. The model has been applied to the Nb-W alloy system and shown to be effective for three representative tasks. (1)

In inverse design, the generative model is coupled with an active learning loop using a physically meaningful curve-level metric—the expected improvement of stress-strain response. Applying this approach, we synthesized the alloy Nb-6W-2Mo-1Zr-6Ta-1.5Hf-0.05C-0.2N (wt %) which, under optimized full-recrystallization annealing conditions, achieves over 35 % improvement in strength without sacrificing ductility compared to commercial Nb521 alloy. (2) The incorporation of a Mixture-of-Experts (MoE) mechanism enables interpretable decomposition of deformation regimes within the stress-strain curves. Expert-specific attention highlights the positive contributions of Ta and Hf to ductility, which are further confirmed by experimental observation and DFT calculation. (3) The generated stress-strain curves are seamlessly integrated into finite element simulations, enabling high-fidelity structural performance prediction under complex loading scenarios.

These results highlight the potential of sequential generative modeling to accelerate the exploration of composition-process-property relationships in refractory alloys. While this study focuses on Nb-W systems and composition-driven descriptors, the framework is readily extendable to other material classes and to processing path modeling, provided sufficient multimodal data are available. This work establishes a unified, mechanism-aware workflow for data-efficient constitutive modeling and design of refractory alloys.

CRediT authorship contribution statement

Daokuo Wei: Visualization, Methodology, Investigation, Formal analysis, Data curation. **Xinyi Pang:** Methodology, Investigation, Data curation. **Jing Wang:** Investigation, Data curation. **Zhishang Li:** Data curation. **Hongxiang Zong:** Writing – original draft, Supervision, Formal analysis, Conceptualization. **Xiangdong Ding:** Supervision, Resources, Project administration. **Turab Lookman:** Writing – review & editing. **Jun Sun:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.actamat.2026.122123.

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